



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PEC- CS801 B/PECIT801B Cryptography & Network Security

UPID : 008257

CS/B.TECH(N)/EVEN/SEM-8/8257/2024-2025/1010

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What is Confidentiality?
- (ii) What is Authentication?
- (iii) What is Public Key Cryptography?
- (iv) What is Message Digest?
- (v) What is Trojan Horse?
- (vi) What is the full form of CIA?
- (vii) What do you mean by cipher text?
- (viii) What is Transposition in Encryption?
- (ix) What is hash value?
- (x) What is Password?
- (xi) What is Adware?
- (xii) What is Worm in computer?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

- 2. Differentiate passive attack from active attack with example. [5]
- 3. Explain Firewall. [5]
- 4. Why Security is required in Cyber media ? [5]
- 5. Differentiate between Symmetric Key and Asymmetric Key Cryptography. [5]
- 6. Explain Digital Certificate. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

- 7. (a) Explain DES algorithm with example. [8]
(b) Compare DES with 3-DES. [7]
- 8. (a) Explain RSA algorithm with a suitable example. [8]
(b) What is Digital Signature? [5]
(c) What is message digest? [2]
- 9. (a) Explain Handshaking protocol of Secure Socket Layer. [8]
(b) Explain different biometric authentication techniques. [7]
- 10. (a) What are the services provided by PGP ? [5]
(b) Explain how PGP provides confidentiality and authentication with suitable diagrams. [8]
(c) What is S/MIME? [2]
- 11. (a) What is firewall? [2]
(b) Explain the working principal of a packet filter firewall with a suitable example. [7]
(c) What are the limitations of firewall? [2]
(d) How packet filter prevents ip spoofing attacks? [4]

*** END OF PAPER ***